ANGELO S. TARZONA

Email: dtarzona@gatech.edu

PROFILE

Fifth-year Ph.D. Candidate at Georgia Institute of Technology with experience coordinating undergraduate instruction and laboratory courses, and a demonstrated commitment to student-centered teaching and instructional development.

EDUCATION

Ph.D. Candidate in Geophysics concentrating in Cryosphere — May 2021 – August 2026
Ph.D. Minor in Geospatial and Environmental Planning
Georgia Institute of Technology, Atlanta, GA
GPA: 3.62/4.00
Advisor: Dr. Winnie Chu
Key Courses: Glacier Dynamics, Environmental Data Analysis, Land Remote Sensing, Geodynamics, Climate Global Change, Climate Change and City Planning, and Geovisualization and Geovisual Analytics

Bachelor of Science in Earth Sciences concentrating in Environmental Geosciences — May 2021
Dickinson College, Carlisle, PA
GPA: 3.49/4.00
Advisor: Dr. Jorden L. Hayes
Key Courses: Environmental and Global Geophysics, Arctic Studies, Structural Geology, Surface Processes, Earth Materials, Field Geology, Sedimentology and Stratigraphy, and Geographic Information Systems

SELECTED DEPARTMENT SERVICES

Committee Member, School of Earth and Atmospheric Sciences Chair Search — Spring 2024
Department Representative, College of Sciences Student Wellness Stakeholder Meeting — Spring 2024
President, Graduate Students in School of Earth and Atmospheric Sciences — Fall 2023 - Spring 2024

SELECTED AWARDS & HONORS

Georgia Institute of Technology
College of Sciences, Earth, and Atmospheric Sciences' Graduate Student Service Award — Spring 2024
College of Sciences Dean's Fellowship — Fall 2021- 2024
Earth & Atmospheric Sciences' Symposium – 6th Place — Spring 2023
College of Sciences, Earth & Atmospheric Sciences' Teacher Assistant of the Year — Spring 2022

RESEARCH EXPERIENCE

Graduate Research Assistant, Georgia Institute of Technology
Fifth-year Ph.D. Candidate — Summer 2021-Present

- Georeferenced ~25,000 archival airborne radar echo sounding points in ArcGIS Pro at Ross Ice Shelf, Antarctica by spatially aligning legacy radar features from this dataset with modern radar observations to enhance spatial accuracy.
- Developed Python-based workflows to semi-automatically extract ice thickness and radar power measurements from georeferenced radar profiles digitized from 35 mm film to enable temporal comparisons with modern observation.
- Collaborated with high school and Georgia Tech students to analyze and reconstruct metadata for this historical dataset as part of an ongoing educational research outreach.

MANUSCRIPTS UNDER PREPARATION

Tarzona, A., Chu, W., Schroeder, D. M., Siegfried, M., Yin, R., Combs, C., Banerjee, R., Nayak, S., and Pellegri, S., (2026). *Recovered Ice Thickness and Basal Reflectivity from Archival Radar Observations over the Ross Ice Shelf, Antarctica*. Journal of Glaciology. Manuscript in preparation.

Tarzona, A., Chu, W., Schroeder, D. M., Siegfried, M., and Amaro, B. (2026). *Echoes of a Frozen Legacy: Reprocessing the Archival Radar Film Record of the Ross Ice Shelf.* Polar Record. Manuscript in Preparation.

CONFERENCE ABSTRACTS

Tarzona, A., Chu, W., Schroeder, D. M., Siegfried, M., Verboncoeur, H., Combs, C., Banerjee, R., and Yin, R. (2025). *Frozen on 35 mm Film: Unlocking 1970s Airborne Radar Echo Sounding Survey in Ross Ice Shelf, Antarctica*. American Geophysical Union (AGU) Fall Meeting, New Orleans, LA, December 15–19, 2025. Abstract #C51A-02.

Tarzona, A., Chu, W., Amaro, B., Schroeder, D. M., Siegfried, M., Verboncoeur, H., and Yin, R. (2024). *Radiometric Interpretation of Archival Airborne Radio Echo Sounding in Ross Ice Shelf*. American Geophysical Union (AGU) Fall Meeting, Washington, DC, December 9–13, 2024. Abstract #NS33A-1205.

Tarzona, A., Chu, W., Verboncoeur, H., Siegfried, M., Schroeder, D. M., Altaweel, A., Amaro, B., and Tran, K. (2023). *Improved Vertical Calibration of the Historical SPRI-NSF-TUD Airborne Radar Echo Sounding Ice Thickness Measurements at Ross Ice Shelf, Antarctica*. American Geophysical Union (AGU) Fall Meeting, San Francisco, CA, December 11–15, 2023. Abstract #C11D-1077.

Tarzona, A., Chu, W., Verboncoeur, H., Siegfried, M., Schroeder, D. M., Combs, L., Altaweel, A., Prabu, A., and Tran, K. (2022). *Geographical repositioning efforts and vertical calibration of Z-scopes from SPRI-NSF-TUD surveys at Ross Ice Shelf, Antarctica*. American Geophysical Union (AGU) Fall Meeting, Chicago, IL, December 12–16, 2022. Abstract #C45D-1123.

Tarzona, A., Chu, W., Tran, K., Teisberg, T., and Dawson, E. (2021). *Four-decades of Ross Ice Shelf changes: Part 2. Comparison using archival SPRI-NSF-TUD and modern ROSETTA-Ice and NASA/NSF IceBridge radio-echo sounding data*. American Geophysical Union (AGU) Fall Meeting, New Orleans, LA, December 13–17, 2021. Abstract #C25E-0866.

CO-AUTHOR PUBLICATION

Keating, K., O'Connell, K., Mount, G. J., Hayes, J. L., DiBiase, R. A., Brantley, S. L., Donaldson, Y., & **Tarzona, A.**: *A short-term critical zone geophysics program designed to recruit, retain, and foster community amongst 1st and 2nd year undergraduates historically marginalized in geosciences*, Journal of Geoscience Education, https://doi.org/10.1080/10899995.2025.2535029, 2025.

Karlsson, N. B., Schroeder, D. M., Sørensen, L. S., Chu, W., Dall, J., Andersen, N. H., Dobson, R., Mackie, E. J., Köhn, S. J., Steinmetz, J. E., **Tarzona, A. S.**, Teisberg, T. O., and Skou, N.: *A Newly Digitised Ice-penetrating Radar Data Set Acquired over the Greenland Ice Sheet in 1971–1979*, Earth Syst. Sci. Data Discuss. [preprint], in review, 2024.

Schroeder, D. M., A. L. Broome, A. Conger, A. Lynch, E. J. Mackie, and **A. Tarzona.,** (2021), *Radiometric analysis of digitized Z-scope records in archival radar sounding film*, Journal of Glaciology, 1–8, doi:10.1017/jog.2021.130.

EMPLOYMENT

Preservation/Mapping Intern, National Trust for Historic Preservation
African American Cultural Heritage Action Fund (AACHAF), Mapping Project Team Summer 2025
Geocoded and mapped approximately 2,519 Black churches across U.S. using ArcGIS Pro, ArcGIS Online, and Python to support the preservation of African American historic sites.

Alliance for Aquatic Resource Monitoring (ALLARM), Dickinson College
Watershed Coordinator 2018-2021
Analyzed different streams throughout Central Pennsylvania using chemical and biological water quality monitoring techniques, subaquatic vegetation assessment, and rendering site maps in ArcGIS for volunteer monitors.

FIELD CAMPAIGNS

Geophysical Field Methods Field Work, Ground Penetrating Radar, Zzyzx, CA Spring 2022, 2024, 2026
Helheim Glacier Field Work, ApRES Recovery, Greenland Summer 2023
Mt. Tabor Historic Cemetery Field Work, Ground Penetrating Radar, Mt. Tabor, PA, USA Spring 2021
Fort Halifax Archeology Field Work, Ground Penetrating Radar, Ft. Halifax, PA, USA Spring 2021

TEACHING EXPERIENCE

Guest Lecturer, Georgia Institute of Technology

EAS 4814: Geophysical Field Methods Spring 2022, 2024, and 2026
Designed and delivered a crash course on ground-penetrating radar field deployment and data processing to support undergraduate geophysical field investigations in Zzyzx, California.

Field Assistant, Georgia Institute of Technology
EAS 4814: Geophysical Field Methods Spring 2022, 2024, and 2026
Assisted with field logistics for 10 – 15 undergraduate students and supported student-led ground-penetrating radar surveys during undergraduate geophysical field campaigns in Zzyzx, California.

Graduate Teaching Assistant, Georgia Institute of Technology
EAS 1601: Habitable Planets Fall 2021- Summer 2022
Independently managed a hybrid lab section and responsible for helping 22 students per section understand lecture concepts and complete lab activities.

Graduate Teaching Assistant Preparation Class Student, Georgia Institute of Technology
CETL 8000: GTA Preparation Fall 2021
Completed formal graduate teaching preparation focused on student learning strategies, classroom engagement, and effective instructional communication, including the design and delivery of a capstone teaching presentation.

MENTORING

Echo Explorers, Georgia Institute of Technology (with Dr. Winnie Chu)
Founder and Lead Mentor

- Coordinate a research team of high school and undergraduate students working on archival radar recovery.
- Mentor students in data analysis, processing workflows, scientific communication, and professional development.

Students mentored:

Shriya Nayak — Environmental Sciences, Georgia Institute of Technology (Fall 2025 – Present)
Simon Pellegri — Physics, Georgia Institute of Technology (Summer 2025 – Present)
Raina Banerjee — Wheeler High School (Spring 2025–Present)
Rebecca Yin — Woodward Academy (Spring 2024 – Present)
Taylor Bandy — Earth & Atmospheric Sciences, Georgia Institute of Technology (2024)
Abdullah Altaweel — Computer Science Engineering, Georgia Institute of Technology (2022–2023)
Kieran Choi-Slattery — Aerospace Engineering, Georgia Institute of Technology (Summer 2022)
Kiera Tran — Earth & Atmospheric Sciences (2021–2022; now Ph.D. Candidate, Georgia Institute of Technology)

Summer Research Experiences for Undergraduates (REU), Georgia Institute of Technology
Co-mentor with Dr. Winnie Chu

- Provided technical training, weekly check-ins, and feedback on analysis and presentation skills.
- Supported students in translating research outcomes into posters and reports

Students mentored:

Cameron Combs — Earth & Atmospheric Sciences REU (Summer 2025)
Laila Combs — Earth & Atmospheric Sciences REU (Summer 2022)

External Undergraduate Advising, GeoPATHS GNOMES Program a Geophysics Field Camp Experience

- Mentored cohorts of approximately **15 – 20 undergraduate students in 2019 and 2022** through a geophysics field camp at the Shale Hills Critical Zone Observatory, Pennsylvania.
- Delivered hands-on instruction in seismic tomography, electrical resistivity tomography, ground-penetrating radar, and surveying techniques to support student understanding from data collection to processing and interpretation.

SKILLS

Geospatial Tools: ArcGIS Pro, QGIS, Google Earth Engine, Agisoft, Flowmapper, Kepler GL, GeoDa, and Tableau
Geophysical Data Processing: ENVI Software, Geogiga Software, Res2Dinv, ResiPy, GPRPy, and GPR-SLICE
Graphics and Productivity Tools: Adobe Photoshop, Adobe Illustrator, and Microsoft Office 365
Programming Languages: MATLAB, Python, SQL, and HTML